Solutions Manual

Worked solutions to the end-of-chapter exercises, Weeks 1–10. Conceptual answers give the reasoning expected; computational answers show the steps. Sketching and lab exercises describe what a complete response contains.

Week 1 — Functions

1. Domain $\{-1, 0, 2, 5\}$, range $\{3, 0, -3, 1\}$. *Not* a function: the input 2 is paired with both 3 and -3, so one input determines two outputs.

2. (a) A circle fails; a vertical line through its interior meets it twice. (b) A downward parabola passes; every vertical line meets it once. (c) A sideways "S" fails; some vertical lines meet it three times. (d) A single steep line passes. The test checks the defining property of a function: that each input has exactly one output, since a vertical line collects all points sharing one x-value.

3. $f(0) = 3$ and $f(-2) = 2(4) - (-2) + 3 = 8 + 2 + 3 = 13$. For $f(t+1)$,

$$f(t+1) = 2(t+1)^2 - (t+1) + 3 = 2(t^2 + 2t + 1) - t - 1 + 3 = 2t^2 + 3t + 4.$$

4. For example $\{(1, 2), (1, 5)\}$ is not a function, since 1 has two outputs. Changing $(1, 5)$ to $(2, 5)$ gives $\{(1, 2), (2, 5)\}$, a function: the repeated input is removed, so every input now has a single output.

5. Shifting left 4 uses $h = -4$, stretching vertically by 3 uses $A = 3$, and shifting down 1 uses $k = -1$:

$$g(x) = 3(x + 4)^2 - 1.$$

The vertex of x^2 at the origin maps to $(h, k) = (-4, -1)$.

6. Read $g(x) = -\frac{1}{2}|x + 2| + 5$ as $A = -\frac{1}{2}$, $h = -2$, $k = 5$. In order: shift left 2; compress vertically by $\frac{1}{2}$ and reflect across the x-axis (because $A < 0$); shift up 5. The corner of $|x|$ at the origin lands at $(h, k) = (-2, 5)$.

7. Track the point $(9,3)$ through $g(x) = 2\sqrt{x-1}+4$. The input transforms by the inside: $x : 9 \to 9-1 = 8$? No, undo carefully. The inside is $x-1$, so the parent's input 9 corresponds to g's input $9+1 = 10$. The output transforms by the outside: $y : 3 \to 2(3)+4 = 10$. So $(9,3) \mapsto (10,10)$. Confirm directly: $g(10) = 2\sqrt{10-1}+4 = 2(3)+4 = 10$. ✓

8. Constants outside f act on the output and move the graph vertically as expected, so $f(x)+3$ raises every output by 3. Constants inside f act on the input and move the graph horizontally in reverse: $f(x-3)$ reaches the parent's value at input 0 when $x-3=0$, that is at $x=3$, so the graph shifts right. The two "+3" moves differ because one adjusts the output (after the function acts) and the other adjusts the input (before the function acts).

9. The cloud is the data itself: scattered points with no exact rule. A linear fit is a *model*, a clean predictable line laid through the cloud to follow its trend. The two are not equal; the model summarizes and forecasts the data without passing through every point. "Model" names the chosen rule that stands in for the data so one can compute with it.

10. With $A < 0$ the graph reflects across the x-axis (the output sign flips), and with $b < 0$ it reflects across the y-axis (the input sign flips). The two reflections are independent and involve the x-axis and the y-axis respectively.

11. A bowl opening upward with lowest point near $(2,-3)$ suggests the parent $f(x) = x^2$. Start with $A = 1$ (upward, unstretched), $b = 1$, $h = 2$ (vertex x), and $k = -3$ (vertex y). The vertex of the model then sits at $(2,-3)$, matching the cloud's lowest point; A is tuned afterward to match how steeply the cloud rises.

12. *Lab exercise.* Answers vary with the generated cloud. A complete response records the parent family, the four fitted constants, the fit score, and one sentence identifying the hardest constant (commonly b, since horizontal scaling is the least visually obvious of the four).

Week 2 — Lines and Parabolas

1. Any two points on a line determine a right triangle with horizontal leg (run) and vertical leg (rise). Because the line meets every horizontal at one fixed angle, all such triangles have the same angles and are therefore similar. Similar triangles have proportional sides, so the ratio rise/run is identical for every pair of points; that common ratio is the slope.

2. Slope: $m = \dfrac{-6-2}{3-(-1)} = \dfrac{-8}{4} = -2$. Point-slope through $(-1,2)$: $y-2 = -2(x+1)$.

Slope-intercept: $y = -2x - 2 + 2 = -2x$, so $f(x) = -2x$. Check: $f(-1) = 2$ and $f(3) = -6$. ✓

3. Point-slope: $y - 1 = -\frac{2}{3}(x - 4)$. Slope-intercept: $y = -\frac{2}{3}x + \frac{8}{3} + 1 = -\frac{2}{3}x + \frac{11}{3}$. The y-intercept is $\frac{11}{3}$. The x-intercept solves $0 = -\frac{2}{3}x + \frac{11}{3}$, giving $x = \frac{11}{2}$.

4. (a) Through $(2, 5)$ and $(2, -1)$: the run is 0, so the slope is undefined; this is the vertical line $x = 2$, which is *not* a function. (b) Through $(0, 4)$ and $(7, 4)$: slope 0, a function (the horizontal line $y = 4$). (c) Through $(1, 1)$ and $(3, 5)$: slope $\dfrac{5-1}{3-1} = 2$, a function.

5. A parabola is the set of points equidistant from a fixed point (the focus) and a fixed line (the directrix). The vertex is the point of the parabola on the axis; its distance to the focus equals its distance to the directrix by definition. Since both distances are measured along the axis, the vertex sits exactly halfway between focus and directrix.

6. Rays arriving parallel to the axis reflect off the curve and all pass through the focus. Two devices: a satellite dish or solar collector, where incoming parallel rays are concentrated *at* the focus; and a headlight or flashlight reflector, where light from a source *at* the focus is sent out as a parallel beam. The directions are reversed in the two cases.

7. Substitute each point into $f(x) = ax^2 + bx + c$. From $(0, 1)$: $c = 1$. From $(-1, 6)$: $a - b + c = 6$, so $a - b = 5$. From $(2, 3)$: $4a + 2b + c = 3$, so $4a + 2b = 2$, i.e. $2a + b = 1$. Adding $a - b = 5$ and $2a + b = 1$ gives $3a = 6$, so $a = 2$, then $b = -3$. Thus $f(x) = 2x^2 - 3x + 1$. Check (-1): $2 + 3 + 1 = 6$. ✓

8. Factor 3 from the variable terms: $3(x^2 + 4x) + 7$. Half of 4, squared, is 4: $3(x^2 + 4x + 4 - 4) + 7 = 3(x + 2)^2 - 12 + 7 = 3(x + 2)^2 - 5$. The vertex is $(-2, -5)$. By formula, $h = -\dfrac{b}{2a} = -\dfrac{12}{6} = -2$, agreeing.

9. Factor: $x^2 - 2x - 15 = (x - 5)(x + 3)$, roots 5 and -3. Axis of symmetry: $x = \dfrac{5 + (-3)}{2} = 1$. Vertex y-value: $f(1) = 1 - 2 - 15 = -16$, so the vertex is $(1, -16)$. The y-intercept is $f(0) = -15$. A correct sketch is an upward parabola crossing at -3 and 5, with vertex $(1, -16)$ and y-intercept $(0, -15)$.

10. Discriminant $b^2 - 4ac$. (a) $x^2 - 4x + 4$: $16 - 16 = 0$, one (doubled) real root at $x = 2$. (b) $2x^2 + x + 3$: $1 - 24 = -23 < 0$, no real roots. (c) $x^2 - 5x + 2$: $25 - 8 = 17 > 0$, two real roots, $x = \dfrac{5 \pm \sqrt{17}}{2} \approx 4.56$ and 0.44.

11. With roots -4 and 2, $f(x) = a(x+4)(x-2)$. Then $f(0) = a(4)(-2) = -8a = -16$, so $a = 2$. The three forms:

$$\text{factored } 2(x+4)(x-2), \quad \text{standard } 2x^2+4x-16, \quad \text{vertex } 2(x+1)^2-18,$$

the last from $h = \dfrac{-4+2}{2} = -1$ and $k = f(-1) = 2-4-16 = -18$.

12. The vertex sits at $h = -\dfrac{b}{2a}$. The root-sum relation gives $r_1 + r_2 = -\dfrac{b}{a}$, so the average of the roots is $\dfrac{r_1+r_2}{2} = -\dfrac{b}{2a} = h$. Thus the vertex's x-coordinate is exactly the average of the roots. Geometrically, a parabola is symmetric about the vertical line through its vertex, and the two roots lie at equal distances on either side of that axis, so their midpoint is the axis itself.

Week 3 — Polynomials and Their Shape

1. End behavior from the leading term. (a) $3x^4$: even degree, positive lead, both ends up. (b) $-x^3$: odd, negative lead, up on the left and down on the right. (c) $-4x^6$: even, negative lead, both ends down. (d) x^5: odd, positive lead, down on the left and up on the right.

2. For large $|x|$, a higher power outgrows every lower power: x^n eventually dwarfs $x^{n-1}, \ldots, x, 1$. So the leading term decides the ends, and the constant and middle coefficients, attached to lower powers, become negligible there. They shift and bend the middle of the graph but cannot change where the ends go.

3. Roots: -1 (multiplicity 1), 2 (multiplicity 3), 4 (multiplicity 1). Degree $1+3+1 = 5$. The graph crosses at all three, since every multiplicity is odd; at $x = 2$ the crossing flattens against the axis because the multiplicity is 3.

4. Just left of r the factor $(x-r)$ is negative; just right it is positive. If m is odd, $(x-r)^m$ keeps the sign of $(x-r)$ (a negative raised to an odd power stays negative), so it flips sign across r. If m is even, $(x-r)^m \geq 0$ on both sides (an even power is nonnegative), so the sign does not change.

5. Roots $-3, 1, 2$ split the line into four intervals.

interval	$(-\infty,-3)$	$(-3,1)$	$(1,2)$	$(2,\infty)$
test x	-4	0	1.5	3
$x+3$	$-$	$+$	$+$	$+$
$x-1$	$-$	$-$	$+$	$+$
$x-2$	$-$	$-$	$-$	$+$
f	$-$	$+$	$-$	$+$

The sign changes at each root (all simple).

6. For $f(x) = -(x+1)^2(x-3)$ the roots are -1 (multiplicity 2) and 3 (multiplicity 1). Intervals: f is $+$ on $(-\infty, -1)$, $+$ on $(-1, 3)$, and $-$ on $(3, \infty)$. The sign does *not* change at $x = -1$ (even multiplicity: a touch) and *does* change at $x = 3$ (odd multiplicity: a crossing). The overall leading sign is negative because of the -1 in front.

7. $f(x) = (x-1)(x+2)^2$. Degree 3, positive lead: down on the left, up on the right. Roots: cross at 1 (simple), touch at -2 (double). Sign chart: $-$ on $(-\infty, -2)$, $-$ on $(-2, 1)$, $+$ on $(1, \infty)$. The curve rises from the lower left, touches the axis at $x = -2$ and turns back down, continues below to cross up at $x = 1$, then climbs to the upper right.

8. Degree-five, positive lead: down-left to up-right. Roots: simple at -2 and 3 (crossings), triple at 0 (a flattened crossing). At the triple root the graph crosses the axis but flattens noticeably against it, the visual signature of multiplicity three, before continuing through.

9. Even degree with simple roots at $-1, 4$ (multiplicity 1 each) and a double root at 2 gives multiplicities summing to at least $1 + 1 + 2 = 4$, so the lowest possible degree is 4 (even, consistent). With negative lead, both ends point down. The curve comes up from the lower left, crosses at -1, rises, touches at 2 and turns, crosses at 4, and falls to the lower right.

10. The sign changes at -1 (from $+$ to $-$) and at 2 (from $-$ to $+$), so both roots have *odd* multiplicity. The lowest-degree factored form using simple roots is $f(x) = (x+1)(x-2)$ up to a positive constant; its sign pattern is $+, -, +$, matching.

11. Without calculus, the sign chart and roots fix which side of the axis the curve occupies and where it crosses or touches, but not the exact height of the turning point between the roots. Both students' graphs can show the correct shape (touch at $x = 1$, cross at $x = 4$, correct ends) while differing in that height. Calculus, through the derivative, is the tool that pins the turning-point height exactly.

12. *Lab exercise.* Answers vary. A complete response records the factored form, the fit score, and one sentence noting that switching a multiplicity from even to odd turns a touch into a crossing at that root (or vice versa).

Week 4 — Rational Functions

1. Zeros at $x = 4$ and $x = -2$ (numerator roots); vertical asymptotes at $x = 1$ and $x = -5$ (denominator roots). No factor is shared between numerator and denominator, so there are no holes and no cancellation.

2. The factor $(x+2)$ is common, so it cancels, leaving $\dfrac{x-3}{x-5}$ with a hole at $x=-2$. The zero is $x=3$ and the vertical asymptote is $x=5$.

3. Near a numerator root the fraction is (small)/(ordinary), which is small, so the graph sits near the x-axis: a zero. Near a denominator root the fraction is (ordinary)/(small), which is enormous, so the graph races off toward $\pm\infty$ along a vertical line: an asymptote.

4. Horizontal asymptotes. (a) Equal degrees: $y=\frac{2}{1}=2$. (b) Equal degrees: $y=\frac{5}{1}=5$. (c) Numerator degree below denominator: $y=0$. (d) Numerator degree above denominator: no horizontal asymptote.

5. $\dfrac{6x^2+x}{3x^2-12}$. Equal degrees, so $y=\frac{6}{3}=2$. Zeros from $x(6x+1)=0$: $x=0$ and $x=-\frac{1}{6}$. Vertical asymptotes from $3(x^2-4)=0$: $x=2$ and $x=-2$.

6. Dividing x^2-5x+6 by $x-1$: quotient $x-4$, remainder 2, so $f(x)=x-4+\dfrac{2}{x-1}$. The slant asymptote is $y=x-4$.

7. Dividing $2x^2+3x-1$ by $x+2$: quotient $2x-1$, remainder 1, so the slant asymptote is $y=2x-1$. The vertical asymptote is $x=-2$. The zeros solve $2x^2+3x-1=0$: $x=\dfrac{-3\pm\sqrt{17}}{4}\approx 0.28$ and -1.78.

8. When the numerator degree exceeds the denominator degree, the ratio of leading terms grows without bound as $|x|\to\infty$, so the values cannot level off at a finite height; there is no horizontal asymptote. The end-behavior curve is the polynomial quotient from long division, and it is a straight line exactly when the numerator degree is one more than the denominator degree.

9. $\dfrac{x-1}{(x+2)(x-3)}$. Features $-2, 1, 3$.

interval	$(-\infty,-2)$	$(-2,1)$	$(1,3)$	$(3,\infty)$
test x	-3	0	2	4
$x-1$	$-$	$-$	$+$	$+$
$x+2$	$-$	$+$	$+$	$+$
$x-3$	$-$	$-$	$-$	$+$
f	$-$	$+$	$-$	$+$

10. $\dfrac{x^2-4}{x^2-1} = \dfrac{(x-2)(x+2)}{(x-1)(x+1)}$; no common factors, so no holes. Zeros at $x = \pm 2$ (both crossings). Vertical asymptotes at $x = \pm 1$. Equal degrees with leading coefficients 1, so horizontal asymptote $y = 1$. The sign chart over $-2, -1, 1, 2$ places the curve above the axis on the outer intervals (approaching $y = 1$) and produces the escapes beside $x = \pm 1$. The sketch is symmetric: outer branches approach $y = 1$ from above, the central branch passes through ± 2 with a small peak at $(0, 4)$? No: $f(0) = \frac{-4}{-1} = 4$, so the central piece peaks near $(0, 4)$ — adjust: the central region lies between the asymptotes and the curve dips below the axis between the zeros. A correct sketch shows the two outer branches and the behavior near each asymptote, with $f(0) = 4$.

11. Zeros at 0 and 3 need numerator factors $x(x-3)$; vertical asymptotes at -1 and 2 need denominator factors $(x+1)(x-2)$. The horizontal asymptote $y = 1$ requires equal degrees with equal leading coefficients, which $\dfrac{x(x-3)}{(x+1)(x-2)}$ satisfies (both degree 2, leading coefficient 1). So $f(x) = \dfrac{x(x-3)}{(x+1)(x-2)}$.

12. *Lab exercise.* Answers vary. A complete response records the factored form, the degree ratio, the fit score, and one sentence noting that raising the numerator degree past the denominator degree replaces the horizontal asymptote with a slant or polynomial end-behavior curve.

Week 5 — Inverses and Radicals

1. Domains. (a) $\dfrac{x}{x^2-9}$: all reals except $x = \pm 3$. (b) $\sqrt{2x-6}$: need $2x - 6 \geq 0$, so $x \geq 3$. (c) $\sqrt[3]{x+1}$: all reals (odd root). (d) $\dfrac{1}{\sqrt{x-4}}$: need $x - 4 > 0$ (strictly, to avoid both the negative radicand and a zero denominator), so $x > 4$.

2. Ranges. (a) $x^2 - 4$: a square is ≥ 0, so the range is $y \geq -4$. (b) $\sqrt{x} + 2$: $\sqrt{x} \geq 0$, so the range is $y \geq 2$. (c) $-|x|$: $|x| \geq 0$, so $-|x| \leq 0$, range $y \leq 0$.

3. One-to-one by the horizontal line test. (a) $3x + 2$: yes (a non-horizontal line). (b) $x^2 - 1$: no (a horizontal line meets it twice). (c) x^3: yes. (d) $|x|$: no (the V is met twice by horizontals above the vertex).

4. An inverse must send each output back to the single input that produced it. If the function is not one-to-one, some output y comes from two different inputs, and the inverse would have to send y to both — impossible for a function. The shared output is exactly the obstruction.

5. Inverses. (a) $f(x) = 3x - 7$: $f^{-1}(x) = \dfrac{x+7}{3}$, domain all reals. (b) $f(x) = \dfrac{x+2}{5}$: $f^{-1}(x) = 5x - 2$, domain all reals. (c) $f(x) = x^3 + 1$: $f^{-1}(x) = \sqrt[3]{x-1}$, domain all reals.

6. Restricting $f(x) = x^2$ to $x \geq 0$ makes it one-to-one, with inverse $f^{-1}(x) = \sqrt{x}$. Check: $f^{-1}(f(x)) = \sqrt{x^2} = |x| = x$ for $x \geq 0$. The inverse has domain $x \geq 0$ and range $y \geq 0$ (the original domain).

7. $\sqrt[3]{x}$ inverts x^3, which takes every real value including negatives (e.g. $(-2)^3 = -8$), so its inverse is defined for all reals. $\sqrt{x}$ inverts x^2, which is never negative, so there is no real input whose square is negative; the domain is restricted to $x \geq 0$. The difference is the parity of the power being inverted.

8. Sketch: $f(x) = x^2$ on $x \geq 0$ is the right half of the upward parabola, starting at the origin; $f^{-1}(x) = \sqrt{x}$ is the upper half of a sideways parabola, also from the origin. Each is the mirror image of the other across the dashed line $y = x$, since the point (a, a^2) on f corresponds to (a^2, a) on f^{-1}.

9. (a) $\sqrt{2x+3} = x$. Square: $2x + 3 = x^2$, so $x^2 - 2x - 3 = 0$, $(x-3)(x+1) = 0$, candidates $3, -1$. Check: $x = 3$ gives $\sqrt{9} = 3$ ✓; $x = -1$ gives $\sqrt{1} = 1 \neq -1$, rejected. Solution $x = 3$. (b) $\sqrt{x+7} = x + 1$. Square: $x + 7 = x^2 + 2x + 1$, so $x^2 + x - 6 = 0$, $(x+3)(x-2) = 0$, candidates $-3, 2$. Check: $x = 2$ gives $\sqrt{9} = 3 = 2+1$ ✓; $x = -3$ gives $\sqrt{4} = 2 \neq -2$, rejected. Solution $x = 2$.

10. Reflecting across $y = x$ sends a point (a, b) to (b, a). A point on both f and f^{-1} is fixed by this reflection, which happens only when $a = b$, i.e. on the line $y = x$. So any intersection of a function with its inverse lies on that line.

11. $f(x) = \sqrt{x-2} + 1$: domain $x \geq 2$ (radicand ≥ 0), range $y \geq 1$ (the $\sqrt{\ }$ is ≥ 0, shifted up 1). Inverse: set $y = \sqrt{x-2} + 1$, swap, solve: $x - 1 = \sqrt{y-2}$, so $y = (x-1)^2 + 2$, giving $f^{-1}(x) = (x-1)^2 + 2$ with domain $x \geq 1$. The inverse's domain $x \geq 1$ equals the range of f, as required.

12. $f(x) = \dfrac{1}{x}$. Swap and solve: $x = \dfrac{1}{y}$ gives $y = \dfrac{1}{x}$, so $f^{-1}(x) = \dfrac{1}{x} = f(x)$. The function is its own inverse. Its graph is therefore symmetric across the line $y = x$, since reflecting it there returns the same curve.

Week 6 — Exponential Functions

1. (a) $5, 10, 20, 40$: constant ratio 2, exponential $f(x) = 5 \cdot 2^x$. (b) $5, 8, 11, 14$: constant difference 3, linear $f(x) = 3x + 5$. (c) $100, 20, 4$: constant ratio $\frac{1}{5}$, exponential

$f(x) = 100\left(\frac{1}{5}\right)^x$.

2. Both 3^x and 3^{-x} pass through $(0, 1)$. The graph of 3^x rises to the right (growth); 3^{-x} is its mirror across the y-axis, falling to the right (decay). Each has the x-axis ($y = 0$) as a horizontal asymptote.

3. (a) $5^{-x} = \left(\frac{1}{5}\right)^x$, decay. (b) $\left(\frac{1}{4}\right)^x$ is already positive-base positive-exponent decay. (c) $2^{-3x} = \left(\frac{1}{8}\right)^x$, since $2^{-3} = \frac{1}{8}$, decay.

4. Each finer compounding adds a little more growth, but each added amount is smaller than the last, so the running total is trapped beneath a fixed ceiling rather than increasing without bound. It approaches $e \approx 2.71828$.

5. $A(10) = 1000\,e^{0.04\cdot 10} = 1000\,e^{0.4} \approx 1000(1.49182) \approx \$1491.82$.

6. $4^x = e^{(\ln 4)x}$ with $\ln 4 \approx 1.3863$, so $f(x) \approx e^{1.3863\,x}$.

7. $0.8^t = e^{(\ln 0.8)t}$ with $\ln 0.8 \approx -0.2231$. The continuous rate is negative, confirming decay: the quantity shrinks at about 22.3% continuous rate per unit time.

8. Each 10 years is one half-life: $64 \to 32$ ($t = 10$), $32 \to 16$ ($t = 20$), $16 \to 8$ ($t = 30$). Remaining masses: 32, 16, 8 grams.

9. $M(t) = M_0\left(\frac{1}{2}\right)^{t/1600}$. After 4800 years, $t/T = 4800/1600 = 3$, so $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$ remains.

10. $\left(\frac{1}{2}\right)^n = \frac{1}{8}$ gives $n = 3$ half-lives. With $T = 30$ years, the age is $3 \times 30 = 90$ years.

11. A radioactive element loses a fixed *fraction* of its mass in each fixed time span (half in each half-life), which is constant-ratio behavior, the defining property of an exponential function. Subtracting a fixed amount each year would be linear and would reach zero in finite time, which decay never does.

12. After 8 years: the 2-year isotope has passed $8/2 = 4$ half-lives, leaving $\left(\frac{1}{2}\right)^4 = \frac{1}{16}$; the 4-year isotope has passed $8/4 = 2$ half-lives, leaving $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$. The shorter-half-life isotope has decayed further, because it halves more often in the same time.

Week 7 — Logarithms

1. (a) $\log_3 81 = 4$ ($3^4 = 81$). (b) $\log_{10} 0.01 = -2$ ($10^{-2} = 0.01$). (c) $\log_2 \frac{1}{16} = -4$ ($2^{-4} = \frac{1}{16}$). (d) $\log_7 1 = 0$. (e) $\ln e^4 = 4$.

2. (a) $5^3 = 125 \iff \log_5 125 = 3$. (b) $\log_2 32 = 5 \iff 2^5 = 32$. (c) $e^0 = 1 \iff \ln 1 = 0$.

3. The graphs of 2^x and $\log_2 x$ are reflections across $y = x$. The exponential passes through $(0,1)$ with horizontal asymptote $y = 0$; the logarithm passes through $(1,0)$ with vertical asymptote $x = 0$ (the y-axis).

4. (a) $\log_b(x^2yz^3) = 2\log_b x + \log_b y + 3\log_b z$. (b) $\log \dfrac{\sqrt{x}}{y^4} = \frac{1}{2}\log x - 4\log y$. (c) $\ln \dfrac{e^2x}{y} = 2 + \ln x - \ln y$ (since $\ln e^2 = 2$).

5. (a) $\log x + 3\log y = \log(xy^3)$. (b) $2\ln x - \frac{1}{2}\ln y = \ln \dfrac{x^2}{\sqrt{y}}$. (c) $\log_b 6 - \log_b 2 = \log_b 3$.

6. The laws apply to products, quotients, and powers, never to sums; $\log_b(M+N)$ has no expansion. The expression $\log_b M + \log_b N$ equals $\log_b(MN)$, the logarithm of the product, which is a different quantity from $\log_b(M+N)$.

7. By change of base: (a) $\log_2 50 = \dfrac{\ln 50}{\ln 2} \approx 5.6439$. (b) $\log_5 100 = \dfrac{\ln 100}{\ln 5} \approx 2.8614$. (c) $\log_3 7 = \dfrac{\ln 7}{\ln 3} \approx 1.7712$.

8. (a) $2^x = 15$: $x = \dfrac{\ln 15}{\ln 2} \approx 3.9069$. (b) $5^x = 2$: $x = \dfrac{\ln 2}{\ln 5} \approx 0.4307$. (c) $e^x = 10$: $x = \ln 10 \approx 2.3026$.

9. (a) $\log_2 16 = 4$ bits. (b) $\log_2 1000 = \dfrac{\ln 1000}{\ln 2} \approx 9.97$ bits. (c) $\log_2 6 \approx 2.58$ bits.

10. $p = \frac{1}{64}$: information $-\log_2 \frac{1}{64} = \log_2 64 = 6$ bits. A rarer event has smaller p, hence larger $\log_2 \frac{1}{p}$, so its occurrence resolves more uncertainty and carries more information.

11. A password chosen from 2^{40} equally likely strings carries $\log_2 2^{40} = 40$ bits. By the product law, the bits of independent characters add: $\log_2(2^k \cdot 2^m) = \log_2 2^k + \log_2 2^m = k + m$, so each character contributes its own bits to the total.

12. Total outcomes $6 \times 4 = 24$, carrying $\log_2 24 \approx 4.585$ bits. The parts add: $\log_2 6 + \log_2 4 \approx 2.585 + 2 = 4.585$, matching, because the product law of logarithms is the additivity of information across independent sources.

Week 8 — Implicit Curves and Conics

1. On $x^2 + y^2 = 25$? (a) $(0, 5)$: $0 + 25 = 25$ ✓. (b) $(3, 4)$: $9 + 16 = 25$ ✓. (c) $(-4, 3)$: $16 + 9 = 25$ ✓. (d) $(2, 2)$: $4 + 4 = 8 \neq 25$, not on the circle.

2. Solving $x^2 + y^2 = 9$ gives $y = \sqrt{9 - x^2}$ (upper semicircle) and $y = -\sqrt{9 - x^2}$ (lower semicircle), each with domain $-3 \le x \le 3$. Together they recombine to the full circle, since every point of the circle has y either nonnegative (upper) or nonpositive (lower).

3. For $y^2 = x^3 - x$: at $x = -1, 0, 1$ the right side is 0, so $y = 0$, giving the points $(-1, 0)$, $(0, 0)$, $(1, 0)$. At $x = \frac{1}{4}$, $x^3 - x = \frac{1}{64} - \frac{1}{4} < 0$, so $y^2 < 0$ is impossible: no real point there.

4. Wherever $x^3 - x > 0$, the equation gives $y = \pm\sqrt{x^3 - x}$, two opposite values, so a vertical line meets the curve twice and it fails the vertical line test. The real locus has two pieces: a bounded oval over $-1 \le x \le 0$ and an unbounded branch over $x \ge 1$, with a gap on $(0, 1)$ where $x^3 - x < 0$.

5. At $x = 3$: $y^2 = 27 - 3 = 24$, so $y = \pm\sqrt{24} = \pm 2\sqrt{6}$. The points are $(3, 2\sqrt{6})$ and $(3, -2\sqrt{6})$. Check: $(2\sqrt{6})^2 = 24 = 3^3 - 3$ ✓.

6. The radius is the distance from the origin to $(6, 8)$: $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$. The circle is $x^2 + y^2 = 100$.

7. $\dfrac{x^2}{25} + \dfrac{y^2}{16} = 1$: $a^2 = 25$, $b^2 = 16$, so $a = 5$, $b = 4$. Intercepts $(\pm 5, 0)$ and $(0, \pm 4)$. The sketch is an ellipse stretching to ± 5 horizontally and ± 4 vertically.

8. $\dfrac{x^2}{9} - \dfrac{y^2}{4} = 1$: $a = 3$, $b = 2$. Vertices $(\pm 3, 0)$; asymptotes $y = \pm\dfrac{b}{a}x = \pm\dfrac{2}{3}x$. The two branches open left and right, hugging the asymptotes.

9. Classify. (a) $3x^2 + 3y^2 = 12$: equal coefficients, circle. (b) $4x^2 + y^2 = 16$: unequal positive, ellipse. (c) $y^2 - x^2 = 1$: opposite signs, hyperbola. (d) $x - 2y^2 = 0$: one squared term, parabola.

10. For large x, the constant 1 is negligible beside the large squared terms, so $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ approaches $\dfrac{x^2}{a^2} = \dfrac{y^2}{b^2}$, which rearranges to $y = \pm\dfrac{b}{a}x$. The branches approach these asymptote lines ever more closely as x grows.

11. Dividing x by a and y by b before squaring replaces $x^2+y^2=1$ with $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$. This stretches the unit circle by a factor a horizontally and b vertically: the curve now reaches $\pm a$ along the x-axis and $\pm b$ along the y-axis, so a and b are the semi-axes.

12. $4x^2-y^2=1$, i.e. $\dfrac{x^2}{1/4}-\dfrac{y^2}{1}=1$. Opposite signs: a hyperbola. Here $a^2=\frac{1}{4}$ so $a=\frac{1}{2}$, vertices $\left(\pm\frac{1}{2},0\right)$; $b=1$, asymptotes $y=\pm\dfrac{b}{a}x=\pm 2x$.

Week 9 — Systems of Linear Equations

1. (a) Two lines with different slopes cross at one point: a unique solution. (b) Two distinct lines with equal slope are parallel and never meet: no solution. (c) One line written two ways is a single line: every point solves both, infinitely many solutions.

2. From the first, $y=5-x$. Substitute: $2x-(5-x)=1$, so $3x=6$, $x=2$, then $y=3$. Check: $2+3=5$ and $2(2)-3=1$ ✓. Solution $(2,3)$.

3. From the first, $y=3x-2$. Substitute into $6x-2y=10$: $6x-2(3x-2)=10$ gives $4=10$, a contradiction. No solution; the lines are parallel.

4. The second equation is -2 times the first ($-2x+4y=-8$ is $x-2y=4$ scaled), so it adds nothing. Substituting $x=4+2y$ into the second yields $-8=-8$, always true. Infinitely many solutions: with $y=t$, the set is $(4+2t,\,t)$.

5. A linear system's solution set is the intersection of lines or planes, which are flat objects. Flat objects intersect in a flat object: a point, the empty set, or an entire line or plane. "Exactly two points" is not a flat set, so it cannot be a solution set.

6. Two arrangements giving no solution: (i) two of the three planes are parallel and distinct, so they share no point; (ii) the three planes form a triangular prism, each pair meeting in a line but with the three lines parallel and distinct, so no point lies on all three.

7. Back-substitute. $4z=8$ gives $z=2$. Then $y-z=1$ gives $y=3$. Then $x-y+2z=5$ gives $x=5+3-4=4$. Solution $(4,3,2)$.

8. Eliminate. R_2-2R_1: $-y-3z=-5$. R_3-R_1: $-3y+2z=7$. From the first of these, $y=5-3z$; substitute: $-3(5-3z)+2z=7$, so $-15+11z=7$, $z=2$, then $y=-1$, then $x=4-(-1)-2=3$. Solution $(3,-1,2)$. Check: $3-1+2=4$ ✓.

9. The second equation is twice the first, so it is redundant. The system reduces to $x + y + z = 2$ and $x - y + z = 0$. Subtracting, $2y = 2$, so $y = 1$; then $x + z = 1$. With $z = t$ free, $x = 1 - t$, giving solutions $(1 - t,\ 1,\ t)$: infinitely many, a line where the two distinct planes meet.

10. Let a, c be adult and child tickets. $a + c = 200$ and $12a + 8c = 2000$. From the first, $c = 200 - a$; substitute: $12a + 8(200 - a) = 2000$, so $4a + 1600 = 2000$, $a = 100$, then $c = 100$. One hundred of each.

11. Let x, y, z be pounds at \$8, \$10, \$14. Total $x + y + z = 10$; value $8x + 10y + 14z = 11(10) = 110$; equal cheaper beans $x = y$. Substituting $x = y$: $2x + z = 10$ and $18x + 14z = 110$. From the first $z = 10 - 2x$; substitute: $18x + 14(10 - 2x) = 110$, so $-10x + 140 = 110$, $x = 3$, then $y = 3$, $z = 4$. Three pounds each of the \$8 and \$10 beans and four of the \$14.

12. Adding a multiple of one equation to another produces a new equation that holds exactly when both originals do, and the step is reversible (subtract the same multiple), so the solution set is unchanged. Scaling by zero is forbidden because it replaces an equation with $0 = 0$, discarding its information and potentially enlarging the solution set.

Week 10 — Matrices and Gaussian Elimination

1.

$$\left[\begin{array}{ccc|c} 3 & -1 & 2 & 5 \\ 1 & 4 & -1 & 0 \\ 2 & 1 & 1 & 7 \end{array}\right]$$

The first column holds the x-coefficients, the second the y-coefficients, the third the z-coefficients, and the column after the bar the constants.

2. (i) *Swap* two rows — reorder the equations. (ii) *Scale* a row by a nonzero constant — multiply an equation through by that constant. (iii) *Replace* a row by itself plus a multiple of another row — add a multiple of one equation to another. Each leaves the solution set unchanged.

3. Replace R_2 by $R_2 - 2R_1$:

$$\left[\begin{array}{cc|c} 1 & 3 & 5 \\ 2 & 1 & 4 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & -5 & -6 \end{array}\right].$$

4. $R_2 - 3R_1$ gives $\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -5 & -10 \end{array}\right]$. The bottom row reads $-5y = -10$, so $y = 2$; then $x + 2(2) = 5$, $x = 1$. Solution $(1, 2)$.

5. $R_2 - 2R_1$: $-y - 3z = -5$. $R_3 - R_1$: $-3y + 2z = 7$. Eliminating y (e.g. $R_3 - 3R_2$ on these) gives $11z = 22$, so $z = 2$; then $y = -1$ and $x = 3$. Solution $(3, -1, 2)$. Check: $3 - 1 + 2 = 4$ ✓.

6. $R_2 + 3R_1$ gives $\left[\begin{array}{cc|c} 1 & -2 & 3 \\ 0 & 0 & 14 \end{array}\right]$. The bottom row reads $0 = 14$, a contradiction, so there is no solution; the row of zero coefficients with nonzero constant reveals it.

7. $R_2 - \frac{3}{2}R_1$ (or $2R_2 - 3R_1$) gives a bottom row of all zeros, $0 = 0$. The variable y is free; from $2x + 4y = 6$, $x = 3 - 2y$. With $y = t$, the solution set is $(3 - 2t,\ t)$.

8. A leading entry in every variable column pins down each variable: unique solution. A row $[0\ \cdots\ 0 \mid c]$ with $c \neq 0$ reads $0 = c$, a contradiction: no solution. A row of all zeros leaves a variable without a leading entry, hence free: infinitely many.

9. The row $[0\ 0\ 0 \mid 5]$ reads $0 = 5$, which is false, so the system has no solution.

10. Atom counts: Fe gives $a = 2c$, O gives $2b = 3c$. Take $c = t$: $a = 2t$, $b = \frac{3t}{2}$. The smallest whole numbers use $t = 2$: $a = 4$, $b = 3$, $c = 2$, giving $4\,\mathrm{Fe} + 3\,\mathrm{O_2} \to 2\,\mathrm{Fe_2O_3}$. Check: Fe $4 = 4$, O $6 = 6$.

11. From R_2, $I_1 = 10 - 2I_2$. The junction equation $I_1 - I_2 - I_3 = 0$ becomes $10 - 2I_2 - I_2 - I_3 = 0$, i.e. $3I_2 + I_3 = 10$. With R_3: $2I_2 + 3I_3 = 9$. Solving, $I_2 = 3$, $I_3 = 1$, and $I_1 = 10 - 6 = 4$. Currents $I_1 = 4$, $I_2 = 3$, $I_3 = 1$ amperes. Junction law: $4 = 3 + 1$ ✓.

12. Each row operation on the matrix is exactly an equation operation in disguise: the columns remember which variable each number belongs to, so manipulating rows is the same as manipulating equations, producing the same solution. The matrix form is preferred for large systems because it strips away the repeated variable names, reducing the work to a mechanical, easily automated sequence of arithmetic steps on the coefficient grid.